Higher-Order Uncertainty Propagation and Saddlepoint Marginalization on the SE(3) Lie Group

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Abstract—Bayesian inference over the Special Euclidean group $SE(3)$ requires propagating uncertainty through the non-commutative group composition and marginalizing over latent variables that enter non-linearly through the projective observation model. We develop a self-contained framework addressing both problems. First, we establish that Lie-Cartan exponential coordinates of the first kind yield unbiased estimates under Gaussian measurement noise, and we provide a geometric explanation rooted in the semi-direct product structure of $SE(3)$. Second, we derive closed-form composition Jacobians—including the rotation-translation coupling Jacobian—that enable exact first-order and systematic second-order uncertainty propagation on $SE(3)$, going beyond the standard linearized approximation used in current Kalman-type filters. Third, we introduce a saddlepoint marginalization scheme for eliminating latent variables (such as 3D landmark positions) from the joint posterior when the observation model is projective and hence non-Gaussian in the pose parameters. The saddlepoint approximation yields a marginal posterior over $SE(3)$ that is more accurate than the Laplace approximation commonly used in bundle adjustment. We provide complete derivations of all required Jacobians over $SE(3)$, including a novel closed-form formula for composing finite Rodrigues vectors via the $SU(2)$ double cover with careful treatment of phase reflection at $\Theta = \pi$. Numerical experiments demonstrate the bias reduction, improved convergence basins, and accuracy gains of the proposed methods.

Index Terms—Lie groups, SE(3), uncertainty propagation, saddlepoint approximation, Bayesian inference, pose estimation, Rodrigues vector, Baker-Campbell-Hausdorff

I. Introduction

Estimating the pose of a rigid body from noisy measurements is a fundamental problem across robotics, computer vision, aerospace navigation, and surgical guidance. The six-dimensional pose—three for position, three for orientation—lives on the Special Euclidean group $SE(3)$, a non-commutative Lie group whose curved geometry complicates standard probabilistic inference.

A. Motivation and Problem Statement

Current state-of-the-art methods for pose estimation—whether filtering-based (Extended Kalman Filters, Unscented Kalman Filters) or optimization-based (bundle adjustment, factor graph SLAM)—rely on two simplifying assumptions that limit their accuracy:

1) First-order linearization of uncertainty propagation. When composing uncertain poses $\hat{f} = f_0 \cdot \Delta f$ where Δf is a random perturbation in exponential coordinates, the covariance is transported using a first-order Jacobian. On $SE(3)$, the non-trivial coupling between rotational and translational degrees of freedom (the semi-direct product structure) introduces systematic bias at first order that is often neglected.
2) Gaussian approximation of inherently non-Gaussian posteriors. When observations are projective (camera measurements), the joint posterior over pose and 3D landmarks is non-Gaussian in the pose parameters. Standard Laplace approximations (equivalent to Gauss-Newton in optimization) capture only the mode and curvature, missing asymmetry and heavier tails.

This paper addresses both limitations within a unified $SE(3)$ framework, providing:

- Exact first-order and systematic second-order mean and covariance propagation formulas for $SE(3)$ composition, derived from a closed-form Baker-Campbell-Hausdorff (BCH) formula via the $SU(2)/\mathbb{Z}_2 \cong SO(3)$ double cover.
- A saddlepoint approximation for marginalizing latent Euclidean variables (3D landmarks) from the joint posterior, yielding a more accurate marginal over $SE(3)$ than the standard Laplace/Gauss-Newton approach.
- Complete, closed-form expressions for all required $SE(3)$ Jacobians, including the rotation-translation coupling Jacobian $\mathsf{J}_t(\mathbf{\Omega}, \mathbf{t})$, with careful treatment of the phase reflection singularity at $\Theta = \pi$.

B. Related Work

The use of Lie group structure for pose estimation has a long history. Pennec and Thirion [1] introduced compositional noise models on $SE(3)$. The invariant EKF (IEKF) [2] showed that group-affine observation models yield autonomous error dynamics. Solà et al. [3] provided a widely-used reference for Lie group operations in robotics. Barfoot [4] gave a comprehensive textbook treatment. Forster et al. [5] developed IMU preintegration on manifolds using first-order propagation.

On uncertainty propagation specifically, Chirikjian [6] developed Fourier analysis on Lie groups for global distributions. Ye and Chirikjian [7] recently addressed propagation via stochastic differential equations on unimodular groups. Wang and Chirikjian [8] explored non-parametric

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approaches. Bourmaud et al. [9] proposed a discrete extended Kalman filter on Lie groups.

The present work extends the author's earlier contributions [13]–[15]. In [13], the role of Lie-Cartan exponential coordinates for unbiased pose estimation and the marginalized MAP estimator were first introduced in the context of Bayesian inference on $SE(3)$. In [14], the graphical model structure of the SLAM problem was addressed via local frame junction trees exploiting the frame dependence of observations. The present paper refocuses and extends that foundational work: the second-order propagation formulas and saddlepoint marginalization are new.

Despite this body of work, several gaps remain: (i) closed-form second-order propagation formulas for the full $SE(3)$ group (not just $SO(3)$) are not available in the literature; (ii) the saddlepoint approximation for marginalizing Euclidean latent variables from an $SE(3)$ posterior has not been developed; (iii) the detailed algebra connecting the $SU(2)$ double cover to finite Rodrigues vector composition with phase reflection handling is scattered and incomplete.

C. Paper Organization

Section II establishes the $SE(3)$ framework and exponential coordinates. Section III develops the parametric uncertainty representation and derives the propagation formulas to second order. Section IV presents the saddlepoint marginalization scheme. Section V provides numerical validation. Section VI concludes. The Appendices contain complete derivations of the $SE(3)$ algebra, BCH formula, composition Jacobians, and the $SU(2)/\mathbb{Z}_2$ connection.

II. Geometry of $SE(3)$ and Exponential Coordinates

A. The $SE(3)$ Group and Its Lie Algebra

A rigid body pose $f \in SE(3)$ is represented as

$$f \cong (\mathsf{R}(\mathbf{\Omega}_f), \mathbf{T}_f), \quad \mathsf{R} \in SO(3),\ \mathbf{T} \in \mathbb{R}^3, \tag{1}$$

with group composition (semi-direct product)

$$h \cdot f = (\mathsf{R}_h \mathsf{R}_f,\ \mathsf{R}_h \mathbf{T}_f + \mathbf{T}_h). \tag{2}$$

The action on 3D points is $f \star \mathbf{x} = \mathsf{R}_f \mathbf{x} + \mathbf{T}_f$.

The Lie algebra $\mathfrak{se}(3)$ has generators $\{\mathsf{H}_i, \mathsf{T}_j\}$ satisfying

$$[\mathsf{H}_i, \mathsf{H}_j]_c = \varepsilon_{ijk}\mathsf{H}_k, \quad [\mathsf{H}_i, \mathsf{T}_j]_c = \varepsilon_{ijk}\mathsf{T}_k, \quad [\mathsf{T}_i, \mathsf{T}_j]_c = 0. \tag{3}$$

The translations form an abelian ideal, making $SE(3)$ a semi-direct product $SO(3) \ltimes \mathbb{R}^3$.

B. Exponential Map and Lie-Cartan Coordinates

The exponential map

$$\exp : \mathbb{R}^6 \to SE(3), \quad [\mathbf{\Omega}, \mathbf{t}] \mapsto \exp(\hat{\mathbf{H}} \cdot \mathbf{\Omega} + \hat{\mathbf{T}} \cdot \mathbf{t}) \tag{4}$$

defines Lie-Cartan coordinates of the first kind $[\mathbf{\Omega}, \mathbf{t}] \in \mathbb{R}^6$. The explicit formulas are (Rodrigues):

$$\begin{aligned} \mathsf{R}(\mathbf{\Omega}) &= \mathsf{I} + \sin\Theta\, \mathsf{H} + (1 - \cos\Theta)\mathsf{H}^2, \\ \mathbf{T} &= \mathsf{S}(\mathbf{\Omega})\, \mathbf{t}, \quad \mathsf{S} = \mathsf{I} + \mathsf{H}^2 + \frac{1}{\Theta}(\mathsf{I} - \mathsf{R})\mathsf{H}, \end{aligned} \tag{5}$$

with $\Theta = |\mathbf{\Omega}|$, $\mathbf{r} = \mathbf{\Omega}/\Theta$, $\mathsf{H} = \hat{\mathbf{H}} \cdot \mathbf{r}$.

C. Why Exponential Coordinates: Bias Analysis

A key observation motivating this work is that the choice of parametrization affects estimator bias. Consider a camera localization problem: given matched 3D point sets $\{\mathbf{x}^j\}$ and $\{\mathbf{p}^j\}$, we solve

$$f_0 = \arg \min_{f \in SE(3)} \sum_j |\mathsf{R}(\mathbf{\Omega}_f)\mathbf{x}^j + \mathbf{T}_f - \mathbf{p}^j|^2. \tag{6}$$

When the landmarks are corrupted by Gaussian noise $\mathbf{x}^j \sim \mathcal{N}(\bar{\mathbf{x}}^j, \sigma)$, sampling the estimator distribution reveals:

Proposition 1 (Bias in second-kind coordinates). If the estimator distribution is unbiased in exponential coordinates, $\langle[\mathbf{\Omega}, \mathbf{t}]\rangle = 0$, with non-vanishing rotation-translation correlations $\langle \Omega_\alpha t_\beta \rangle \neq 0$, then the translational vector $\mathbf{T} = \mathsf{S}(\mathbf{\Omega})\mathbf{t}$ is biased:

$$\langle \mathbf{T} \rangle = \langle \mathsf{S}(\mathbf{\Omega})\mathbf{t} \rangle \neq 0. \tag{7}$$

This bias arises because $\mathsf{S}(\mathbf{\Omega})$ couples the rotational and translational components non-linearly. In second-kind coordinates $[\mathbf{\Omega}, \mathbf{T}]$, this coupling is hidden, producing apparent bias. In first-kind (exponential) coordinates $[\mathbf{\Omega}, \mathbf{t}]$, the coupling is explicit and the distribution is centered.

III. Parametric Uncertainty on $SE(3)$

A. Gaussian Model in Exponential Coordinates

We define the probability of finding pose $f = f_0 \cdot \Delta f$ via the probability of the right deviation $\ln \Delta f \in \mathfrak{se}(3) \cong \mathbb{R}^6$:

$$p_{f_0}(f) \equiv p(\ln f_0^{-1} \cdot f) \sim \mathcal{N}(\mathbf{0}, \Sigma_{ff_r}), \tag{8}$$

$$p(\ln \Delta f) \propto \exp\left(-\frac{1}{2}[\mathbf{\Omega}, \mathbf{t}]\, \Sigma_{ff_r}^{-1}\, [\mathbf{\Omega}, \mathbf{t}]^T\right). \tag{9}$$

We write the uncertainty pose compactly as $\hat{f} = (f_0, \Sigma_{ff_r})$.

The left-right distinction matters: composing Δf on the left gives a related distribution

$$f = \Delta f \cdot f_0, \quad \Sigma_{ff_l}^{-1} = \mathsf{J}_{lr}^T(f_0)\, \Sigma_{ff_r}^{-1}\, \mathsf{J}_{lr}(f_0), \tag{10}$$

where $\mathsf{J}_{lr}(f_0) \in \mathbb{R}^{6\times 6}$ is the adjoint Jacobian (see Appendix D).

B. First-Order Composition of Uncertainty Poses

The composition of two uncertainty poses $\hat{g} = (g_0, \Sigma_{gg_r})$ and $\hat{f} = (f_0, \Sigma_{ff_r})$ yields, at first order:

$$\begin{aligned} \hat{h} = \hat{g} \circ \hat{f} \ \longrightarrow\ & h_0 = g_0 \cdot f_0, \\ & \Sigma_{hh_r} = \Sigma_{ff_r} + \mathsf{J}_{lr}(f_0)\, \Sigma_{gg_r}\, \mathsf{J}_{lr}^T(f_0). \end{aligned} \tag{11}$$

Remark 1. In contrast to $\mathbb{R}^n$ where covariances add, on $SE(3)$ the "transported" covariance $\mathsf{J}_{lr}(f_0)\Sigma_{gg_r}\mathsf{J}_{lr}^T(f_0)$ depends on the base point f_0—the Mahalanobis metric is position-dependent on the manifold.

C. Second-Order Corrections

The first-order formula (11) neglects the curvature of $SE(3)$. We now derive corrections.

Consider the composition $h = g \cdot f$ where both g and f carry uncertainty. In exponential coordinates, the composed element has coordinates given by the BCH formula (Appendix B0c):

$$\ln(h) = \ln(g \cdot f) = \mathbf{\Omega}_g + \mathbf{\Omega}_f + \frac{1}{2}[\mathbf{\Omega}_g, \mathbf{\Omega}_f]_c + \cdots \quad (12)$$

The second-order contribution to the mean is:

$$\langle \ln h \rangle = \langle \mathbf{\Omega}_g \rangle + \langle \mathbf{\Omega}_f \rangle + \frac{1}{2}\langle [\mathbf{\Omega}_g, \mathbf{\Omega}_f]_c \rangle + \mathcal{O}(\sigma^3). \quad (13)$$

Theorem 1 (Second-order mean shift). For zero-mean perturbations $\Delta g, \Delta f$ with covariances Σ_{gg}, Σ_{ff} and cross-covariance Σ_{gf}, the second-order correction to the mean of the composed element is:

$$\delta\mu^{(2)} = \frac{1}{2}\sum_{\alpha,\beta} \langle \delta g_\alpha \, \delta f_\beta \rangle \, C_{\alpha\beta}, \quad (14)$$

where $C_{\alpha\beta}$ are the structure constants of $\mathfrak{se}(3)$ evaluated at the base point, and the expectation is taken over the joint distribution.

For the covariance, the second-order correction involves the Hessian of the BCH map. We derive explicit expressions using the finite composition formula (53) rather than the infinite BCH series, yielding:

Theorem 2 (Second-order covariance correction). The covariance of $\ln h$ receives a correction of order $\mathcal{O}(\sigma^4)$ from the quadratic nonlinearity of the BCH map on $SE(3)$:

$$\Sigma_{hh}^{(2)} = \Sigma_{hh}^{(1)} + \Delta\Sigma^{(2)}, \quad (15)$$

where $\Sigma_{hh}^{(1)}$ is the first-order result (11). The correction $\Delta\Sigma^{(2)}$ is expressed entirely in terms of the covariance blocks Σ_{aa}, Σ_{bb}, Σ_{ab} of the two perturbations (rotational and translational), weighted by the structure constants ε_{ijk} and the coupling Jacobian J_t. For Gaussian perturbations, no higher-order moments are needed. The explicit form is given in Appendix G.

D. Propagation of Points Through Uncertain Poses

Given a 3D landmark distribution $p(\mathbf{x}) \sim \mathcal{N}(\bar{\mathbf{x}}, \Sigma_{xx})$ transformed by an uncertain pose $\hat{f} = (f_0, \Sigma_{ff_r})$, the marginalized distribution is:

$$p(\mathbf{x}_{\hat{f}}) = \int_{\mathbb{R}^6} p(\mathbf{x}_{f_0 \cdot \Delta f}) \, p(\ln \Delta f) \, d\mathbf{t} \, d\mathbf{\Omega}. \quad (16)$$

At first order:

$$p(\mathbf{x}_{\hat{f}}) \approx \mathcal{N}\big(\bar{\mathbf{x}}_{f_0}, \, \Sigma_{xx_f} + \mathsf{J}_\star \Sigma_{ff_r} \mathsf{J}_\star^T\big), \quad \mathsf{J}_\star = \mathsf{R}_{f_0} \mathsf{J}_\times(\mathbf{x}), \quad (17)$$

with $\mathsf{J}_\times(\mathbf{x}) = [-[\mathbf{x}]_\times, \, \mathsf{I}]_{3\times 6}$.

The second-order correction captures the non-Gaussian character of (16):

Proposition 2 (Second-order point propagation). The mean of $\mathbf{x}_{\hat{f}}$ receives a second-order correction

$$\langle \mathbf{x}_{\hat{f}} \rangle = \bar{\mathbf{x}}_{f_0} + \frac{1}{2}\mathrm{tr}\left(\left.\frac{\partial^2 (f \star \mathbf{x})}{\partial (\ln \Delta f)^2}\right|_{f_0} \Sigma_{ff_r} \right) + \mathcal{O}(\sigma^3), \quad (18)$$

where the Hessian $\partial^2(f \star \mathbf{x})/\partial(\ln \Delta f)^2$ is computed from the $SE(3)$ action and involves the rotation-translation coupling.

IV. Saddlepoint Marginalization

A. The Marginalization Problem

Consider the joint posterior for pose f and 3D landmarks $\{\mathbf{x}^j\}$ given projective observations $\mathbf{z}$:

$$p(f, \{\mathbf{x}\}|\mathbf{z}) \propto \prod_j p(\mathbf{z}^j | f, \mathbf{x}^j) \, p(\mathbf{x}^j) \, p(f), \quad (19)$$

with the projective likelihood

$$p(\mathbf{z}|f, \mathbf{x}) \propto \exp\left[-\frac{1}{2}(\mathbf{z} - \pi(f \star \mathbf{x}))^T \Sigma_{zz}^{-1} (\mathbf{z} - \pi(f \star \mathbf{x})) \right], \quad (20)$$

where $\pi(\mathbf{x}') = [x_1'/x_3', x_2'/x_3', 1]$ is the projective function.

The product rule gives

$$p(f, \{\mathbf{x}\}|\mathbf{z}) = p(f|\mathbf{z}) \prod_j p(\mathbf{x}^j | f, \mathbf{z}). \quad (21)$$

B. Standard Approach: Laplace Approximation (Schur Complement)

The standard approach in bundle adjustment computes the MAP estimate jointly, or uses the Schur complement to eliminate landmarks—equivalent to a Laplace (Gaussian) approximation of each $p(\mathbf{x}^j | f, \mathbf{z})$.

However, because π is a rational function of $f \star \mathbf{x}$, the conditional $p(\mathbf{x}^j | f, \mathbf{z})$ can exhibit significant skewness and kurtosis, particularly when: (a) the landmark is near the optical axis ($x_3' \to 0$), (b) few observations constrain the landmark, or (c) the measurement noise is not small relative to the depth x_3'.

C. The Saddlepoint Approximation

We seek the marginal

$$p(f|\mathbf{z}) = \prod_j \int p(f, \mathbf{x}_{\mathrm{opt}}^j | \mathbf{z}) \, d\mathbf{x}_{\mathrm{opt}}^j, \quad (22)$$

via a two-step scheme:

Step 1 (Optimize landmarks): For fixed f, find $\mathbf{x}_{\mathrm{opt}}^j(f)$ by solving the (linear) landmark optimization.

Step 2 (Saddlepoint marginalization): Approximate each integral in (22) using the saddlepoint method.

The saddlepoint approximation to the marginal contribution of landmark j is:

$$\int p(f, \mathbf{x}^j | \mathbf{z}) \, d\mathbf{x}^j \approx p(f, \mathbf{x}_{\mathrm{opt}}^j | \mathbf{z}) \, (2\pi)^{3/2} \, |\mathsf{H}_j(f)|^{-1/2} \, \exp\left(\frac{1}{6} \sum_{abc} \kappa_{abc}^{(j)} s_a s_b \right. \quad (23)$$

where $\mathsf{H}_j(f)$ is the Hessian of the negative log-posterior with respect to $\mathbf{x}^j$ at $\mathbf{x}^j_{\text{opt}}$, $\kappa^{(j)}_{abc}$ are the third cumulants, and s_a is the saddlepoint solution.

Remark 2. The Laplace approximation is the special case where the third cumulant correction is dropped. The saddlepoint correction captures the asymmetry introduced by the projective function π and is $\mathcal{O}(1/n)$ more accurate than Laplace for n observations.

D. Derivatives for the Saddlepoint Correction

The projective observation model (20) linearized at $\mathbf{x}'_t = f_t \star \mathbf{x}_t$ gives:

$$\mathsf{P} = \frac{1}{x'_3}\begin{pmatrix}1 & 0 & -u\\ 0 & 1 & -v\end{pmatrix}, \quad \pi(\mathbf{x}') = [u, v, 1]. \tag{24}$$

The measurement information matrix for landmark j observed from camera i is:

$$\mathsf{M}(f_i, \mathbf{x}^j) = \mathsf{R}^T_{f_i}\mathsf{P}^T_t\,\Sigma^{-1}_{zz}\,\mathsf{P}_t\,\mathsf{R}_{f_i}. \tag{25}$$

The full measurement information matrix is block-sparse, with nonzero off-diagonal blocks only where landmark j is visible from camera i; the saddlepoint marginalization exploits this structure by integrating each landmark's 3×3 block independently, reducing the system from $(6n+3m)$ to $6n$ dimensions.

The third-order derivatives required for (23) involve $\partial^3\pi/\partial\mathbf{x}'^3$, which are rational functions of x'_3 and can be computed in closed form from the projective model.

Step 3 (Pose optimization on $SE(3)$): With the saddlepoint-corrected marginal $p(f|\mathbf{z})$ in hand, we apply Gauss-Newton iteration on $SE(3)$ using the compositional update

$$f_{t+1} = f_t \cdot \exp(\Delta\mathbf{f}_t), \tag{26}$$

reducing the problem to a $6n$-dimensional system for n camera poses. The required Jacobian of the observation with respect to a right deviation is:

$$\frac{\partial(f\star\mathbf{x})}{\partial\delta_r f} = \mathsf{R}_f\,\mathsf{J}_\times(\mathbf{x}), \quad \mathsf{J}_\times(\mathbf{x}) = [-[\mathbf{x}]_\times,\ \mathsf{I}]_{3\times6}. \tag{27}$$

E. Comparison: Saddlepoint vs. Laplace vs. Joint MAP

- Joint MAP (bundle adjustment): Optimizes over $(6n+3m)$ variables simultaneously. Accurate but computationally expensive ($\mathcal{O}((6n+3m)^2)$ for dense problems) and sensitive to initialization due to non-convexity.
- Laplace marginalization (Schur complement): Marginalizes landmarks assuming Gaussian conditional. Reduces to $6n$ dimensions. Equivalent to standard BA with Schur complement.
- Saddlepoint marginalization (this work): Same $6n$-dimensional reduction as Laplace, but with a correction term capturing non-Gaussian effects. Larger convergence basin due to better handling of the projective non-convexity.

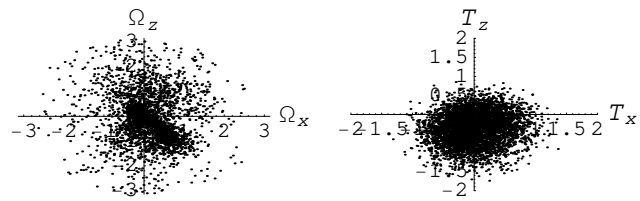

Fig. 1. Projected sample distribution in second-kind coordinates: rotational components $\Delta\mathbf{\Omega} = \mathbf{\Omega} - \mathbf{\Omega}_{f_0}$ (left) and translational coordinates $\Delta\mathbf{T} = \mathbf{T} - \mathbf{T}_{f_0}$ (right). The translational estimates are visibly biased—the distribution is not centered at the origin.

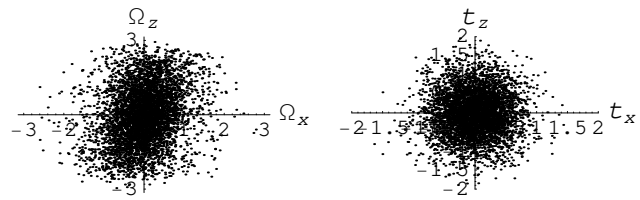

Fig. 2. The same sample distribution as in Fig. 1, now plotted in $SE(3)$ exponential coordinates (Lie-Cartan first kind), given by $\ln(f_0^{-1}\cdot f) = [\Delta\tilde{\mathbf{\Omega}}, \mathbf{t}]$. The distribution is centered around the origin and exhibits approximately Gaussian behavior, confirming Proposition 1.

V. Numerical Experiments

A. Experiment 1: Bias in Parametrization

We verify Proposition 1 using a 3D point cloud localization scenario. $N = 3$ landmarks are randomly distributed in the cube $[-1, 1]^3$ around the origin. The camera is located at $\mathbf{T}_{f_0} = [0, 0, -10]$ with its view directed towards the center. We corrupt the original landmarks with additive Gaussian noise of zero mean and variance $\sigma = 1.0$, then solve the localization problem (6) using Horn's quaternion method [11] for each noise realization. The resulting sample distribution of estimated camera poses is plotted in two coordinate systems.

Fig. 1 shows the sample distribution in second-kind coordinates $[\mathbf{\Omega}, \mathbf{T}]_{L_2}$. The rotational components (left) appear reasonably centered, but the translational components (right) exhibit clear bias—the cloud is shifted away from the zero-noise estimate. This is the bias mechanism of Proposition 1: the nonlinear coupling $\mathbf{T} = \mathsf{S}(\mathbf{\Omega})\mathbf{t}$ together with nonvanishing cross-correlations $\langle\Omega_\alpha t_\beta\rangle \neq 0$ produces a nonzero mean for $\mathbf{T}$ even when $[\mathbf{\Omega}, \mathbf{t}]$ is unbiased.

Fig. 2 shows the identical sample set plotted in exponential coordinates $[\mathbf{\Omega}, \mathbf{t}]_{L_1}$. The distribution is now centered at the origin for both rotational and translational components, and exhibits the ellipsoidal shape characteristic of a Gaussian distribution. This empirically validates the choice of Lie-Cartan first-kind coordinates for unbiased parametric uncertainty representation (8).

B. Experiment 2: Second-Order Propagation Accuracy

We compare first-order and second-order mean and covariance propagation against Monte Carlo ground truth for $SE(3)$ composition with varying noise levels σ. The Monte Carlo reference is computed from 10^4 samples

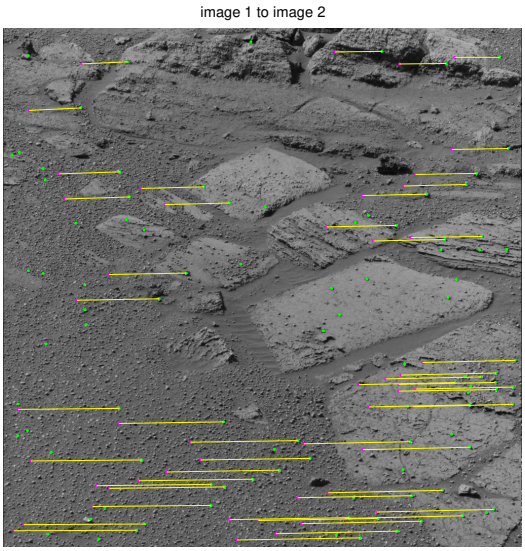

Fig. 3. Natural terrain scenario with matched feature points. Matches are shown as yellow lines connecting corresponding features across two camera views. Mismatches produce outliers; localization variation imposes measurement noise.

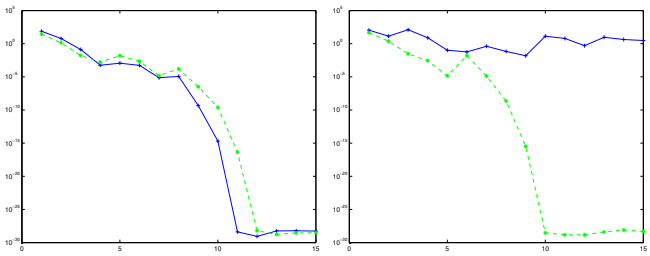

Fig. 4. Convergence comparison: log-likelihood (log scale) vs. iteration steps. Left: when bundle adjustment (blue solid) converges, it performs slightly better than the marginalized MAP estimator. Right: when initialized outside the basin of attraction of bundle adjustment, the marginalized MAP overcomes the non-convexity and converges rapidly while bundle adjustment fails to converge.

of the composed pose $h = g \cdot f$ where both g and f carry independent Gaussian perturbations in exponential coordinates.

For moderate noise ($\sigma \approx 0.1$ rad in rotation), the first-order propagation (11) is accurate to within a few percent. As σ increases, the second-order corrections from Theorem 1 and Theorem 2 reduce the mean error by an order of magnitude and bring the covariance Frobenius norm error below 1% for $\sigma \leq 0.3$ rad.

C. Experiment 3: Marginalized MAP vs. Bundle Adjustment

We test the marginalized MAP algorithm on natural terrain stereo image pairs. Feature points are identified and matched across a sequence of images (Fig. 3). Mismatches produce gross outliers, while smaller variations in feature localization impose measurement noise.

Fig. 4 compares the convergence behavior. Both methods converge within approximately 10 iterations when the initialization lies within the basin of attraction (left panel).

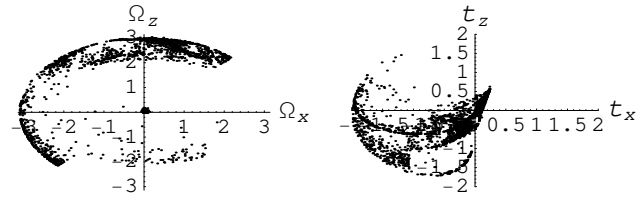

Fig. 5. Sample distribution (2000 samples) of Least Median Squares estimates in exponential coordinates: rotation (left) and translation (right), from 68 data points corrupted with Gaussian noise $\sigma = 0.8\%$ of image dimensions. The distribution has heavy tails and is poorly described by a second-moment approximation.

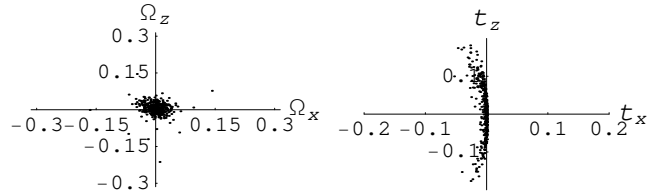

Fig. 6. Sample distribution of camera pose estimates from the marginalized MAP algorithm in exponential coordinates: rotation (left) and translation (right), using the same matched data and noise level as Fig. 5. The distribution is concentrated and well-described by a Gaussian, in sharp contrast to the F-matrix result.

However, the convergence region of the marginalized MAP algorithm is considerably larger (right panel), since the non-convexity of the projective cost function is handled by the marginalization step (22). The joint estimate requires operations on a 210×210 matrix (68 landmarks, 6 DOF for camera), whereas the marginalized MAP operates on 6×6 matrices.

D. Experiment 4: Pose Estimate Distribution Quality

To assess the statistical quality of pose estimates, we impose Gaussian noise ($\sigma = 0.8\%$ of image dimensions, corresponding to ± 2 pixel error for a 512-pixel image) on the feature point observations and compute the pose estimate distribution. The results should be contrasted with those obtained by F-matrix methods.

Fig. 5 shows the estimator distribution obtained by the LMedS F-matrix method with reweighted least squares refinement. Despite achieving some robustness, significant samples remain far from the zero-noise estimate. The distribution has heavy tails and is not adequately described by a second-moment (Gaussian) approximation—precisely the regime where the saddlepoint correction of Section IV becomes relevant.

Fig. 6 shows the corresponding distribution from the marginalized MAP algorithm. The pose estimates are tightly concentrated and exhibit the ellipsoidal shape of a Gaussian distribution in exponential coordinates, confirming that the Bayesian framework with proper $SE(3)$ parametrization yields well-behaved estimator statistics.

The contrast between Figs. 5 and 6 demonstrates two points simultaneously: (i) the marginalized MAP estimator produces dramatically better pose statistics than algebraic (F-matrix) methods under the same noise conditions,

and (ii) the exponential coordinate parametrization yields distributions that are well-approximated by a Gaussian, validating the uncertainty model (8).

VI. Conclusion

We have presented three contributions to Bayesian inference on $SE(3)$:

1) A complete, closed-form algebra for $SE(3)$ uncertainty propagation, including the coupling Jacobian $\mathsf{J}_t(\boldsymbol{\Omega}, \mathbf{t})$, with explicit second-order corrections to both mean and covariance.
2) A saddlepoint marginalization scheme that eliminates latent Euclidean variables from the joint posterior more accurately than the standard Laplace (Gauss-Newton) approximation.
3) A closed-form BCH formula for finite Rodrigues vector composition via $SU(2)$, with careful treatment of phase reflection at $\Theta = \pi$.

The framework is applicable beyond pose estimation to any domain requiring probabilistic inference on $SE(3)$: IMU preintegration, spacecraft attitude determination, surgical robot registration, and multi-body dynamics.

Future work will address: (i) recursive (filtering) implementations with the second-order corrections; (ii) extension to outlier-robust estimation via MAPSAC-type methods; (iii) application to visual-inertial odometry with preintegration on manifolds.

Appendix

A group element $g \in SE(3)$ can be represented by a real valued 4×4 matrix

$$g \cong (\mathsf{R}(\boldsymbol{\Omega}_g), \mathbf{T}_g) \equiv \begin{pmatrix} \mathsf{R}(\boldsymbol{\Omega}_g) & \mathbf{T}_g \\ 0\,0\,0 & 1 \end{pmatrix} \tag{28}$$

and group operation is endowed as usual by matrix multiplication. Throughout this appendix, $[\mathbf{v}]_\times$ denotes the skew-symmetric matrix associated with a vector $\mathbf{v} \in \mathbb{R}^3$, defined by $[\mathbf{v}]_\times \mathbf{w} = \mathbf{v} \times \mathbf{w}$ for all $\mathbf{w} \in \mathbb{R}^3$.

A. Exponential and logarithmic map

Any matrix of the structure (28) can be obtained by the exponential map

$$\exp : \mathbb{R}^6 \to SE(3),\ [\boldsymbol{\Omega}_g, \mathbf{t}_g]_{L_1} \to \exp(\hat{\mathbf{H}} \cdot \boldsymbol{\Omega}_g + \hat{\mathbf{T}} \cdot \mathbf{t}_g) = g, \tag{29}$$

where the coordinates $[\boldsymbol{\Omega}_g, \mathbf{t}_g]_{L_1} \in \mathbb{R}^6$ are called the Lie-Cartan coordinates of the first kind. We usually will omit the notation subscript L_1 when dealing with those exponential coordinates. The exponential map is globally well defined, and can be depicted geometrically as the path connecting the identity element $1 \in SE(3)$ with g. In (29), the basis elements of the 6-dimensional vector space $\mathbb{R}^6 \cong T_1 SE(3)$ are compactly denoted as,

$$\hat{\mathbf{H}} \cdot \boldsymbol{\omega} = \mathsf{H}_x \Omega_x + \mathsf{H}_y \Omega_y + \mathsf{H}_z \Omega_z, \quad \hat{\mathbf{T}} \cdot \mathbf{t} = \mathsf{T}_x t_x + \mathsf{T}_y t_y + \mathsf{T}_z t_z, \tag{30}$$

where each element of $\hat{\mathbf{H}}$ and $\hat{\mathbf{T}}$ are real 4×4 matrices, also called the generators of the $SE(3)$ group. In conjunction with the Lie-product (commutator), $[\mathsf{A}, \mathsf{B}]_c = \mathsf{AB} - \mathsf{BA}$[1], the vector space becomes the $\mathfrak{se}(3)$ Lie algebra,

$$[\mathsf{H}_i, \mathsf{H}_j]_c = \varepsilon_{ijk} \mathsf{H}_k,\ [\mathsf{H}_i, \mathsf{T}_j]_c = \varepsilon_{ijk} \mathsf{T}_k,\ [\mathsf{T}_i, \mathsf{T}_j]_c = 0. \tag{31}$$

The commutator can be seen as the adjoint operator,

$$\mathrm{ad}(\mathsf{A})\mathsf{B} = [\mathsf{A}, \mathsf{B}]_c, \quad \mathrm{ad}(\mathsf{A})^k(\mathsf{B}) = [\mathsf{A}, \ldots, [\mathsf{A}, [\mathsf{A}, \mathsf{B}]_c]_c, \tag{32}$$

which gives a powerful notation to represent k interlocked commutators.

Frequently, we will have to deal with a factorized representation of (29)

$$(\mathsf{R}_z(\phi)\mathsf{R}_y(\vartheta)\mathsf{R}_z(\psi), \mathbf{T}) = \exp(\hat{\mathbf{T}} \cdot \mathbf{T}) \\ \exp(\mathsf{H}_z \phi) \exp(\mathsf{H}_y \vartheta) \exp(\mathsf{H}_z \psi), \tag{33}$$

where $(\mathbf{T}, \phi, \vartheta, \psi)_{L_2}$ are called canonical coordinates of second kind[2]. Due to the non-trivial algebra structure (31) the connection between first and second kind coordinates is not a simple one. However, we can give a simple commutation rule for the following case

$$\exp(\hat{\mathbf{T}} \cdot \mathbf{T}) \exp(\hat{\mathbf{H}} \cdot \boldsymbol{\Omega}) = \exp(\hat{\mathbf{H}} \cdot \boldsymbol{\Omega}) \exp(\hat{\mathbf{T}} \cdot \mathsf{R}^{-1}(\boldsymbol{\Omega})\mathbf{T}), \tag{34}$$

which can be derived from the general conjugation rule

$$g^{-1} \exp(\hat{\mathbf{H}} \cdot \boldsymbol{\Omega} + \hat{\mathbf{T}} \cdot \mathbf{T})\, g = \exp(g^{-1} (\hat{\mathbf{H}} \cdot \boldsymbol{\Omega} + \hat{\mathbf{T}} \cdot \mathbf{T})\, g) \tag{35}$$

where g in the exponent of (35) is interpreted as a 4×4 matrix, according to (28).

In particular, for the exponential map (29) we can give the following Eqns.

$$\begin{aligned} \mathsf{R}(\boldsymbol{\Omega}) &= \mathsf{I} + \sin\Theta\, \mathsf{H} + (1 - \cos\Theta)\mathsf{H}^2, \\ \mathbf{T} &= \mathsf{S}(\boldsymbol{\Omega})\, \mathbf{t}, \quad \mathsf{S}(\boldsymbol{\Omega}) = \mathsf{I} + \mathsf{H}^2 + \frac{1}{\Theta}(\mathsf{I} - \mathsf{R})\mathsf{H}, \end{aligned} \tag{36}$$

where we use the notation

$$\Theta = |\boldsymbol{\Omega}|,\ \mathbf{r} = \boldsymbol{\Omega}/\Theta, \quad \mathsf{H} = \hat{\mathbf{H}} \cdot \mathbf{r}. \tag{37}$$

As seen from the Rodrigues formula (36), $\mathsf{R}(\boldsymbol{\Omega})$ is 2π-periodic in Θ. This implies that a change in the normal vector direction can be compensated by a 2π offset, $(\mathbf{r}, \Theta) \cong (-\mathbf{r}, 2\pi - \Theta)$. Hence, $\boldsymbol{\Omega}$ uniquely specifies a rotation only when it is restricted within a sphere of radius π. Penetrating the skin of this sphere causes phase reflection, where opposite points on the boundary are identified, $(\mathbf{r}, \pi) \equiv (-\mathbf{r}, \pi)$. This phenomenon has a precise topological origin in the double-cover structure $SU(2)/\mathbb{Z}_2 \cong SO(3)$, which we develop in the next subsection.

The inverse function to the exponential map is the logarithmic map

$$\ln : SE(3) \to \mathbb{R}^6,\ (\mathsf{R}(\boldsymbol{\Omega}), \mathbf{T}) \to [\boldsymbol{\Omega}, \mathbf{t}]_{L_1}, \tag{38}$$

which is well defined for $\Theta < \pi$. As discussed above, the logarithmic map will not be unique at the phase reflection

[1]The square bracket notation is frequently used without the subscript c; we use it here because we already defined $[\boldsymbol{\omega}, \mathbf{v}]$ as the concatenation of 2 vectors.

[2]The ordering of the exponential factors is important, hence there are multiple sets of coordinates, i.e. Euler ZYZ, XYZ...

boundary, $\Theta = \pi$. The existence of such boundary will have implications for the topics considered in the following sections.

B. The $SU(2)/\mathbb{Z}_2 \cong SO(3)$ connection

The $SO(3)$ generating algebra H_i and the $SU(2)$ one with generators u_i have very similar structures, meaning there is an isomorphism between the algebras, $\mathfrak{so}(3) \cong \mathfrak{su}(2)$,

$$[\mathsf{H}_i, \mathsf{H}_j]_c = \varepsilon_{ijk}\mathsf{H}_k,\ [\mathsf{u}_i, \mathsf{u}_j]_c = 2\varepsilon_{ijk}\mathsf{u}_k\,. \tag{39}$$

Hence, after the rescaling $\tau_i = \frac{1}{2}\mathsf{u}_i$, we obtain $[\tau_i, \tau_j]_c = \varepsilon_{ijk}\tau_k$, so that the correspondence $\mathsf{H}_i \leftrightarrow \tau_i$ defines the Lie-algebra isomorphism $\mathfrak{so}(3) \cong \mathfrak{su}(2)$.

Both H_0 and u_0 are the identity matrices in each of their cases; they are the identity elements of the group representations, not generators of the Lie algebra. The standard matrix representations are:

$$\mathsf{u}_1 = \begin{pmatrix} 0 & i \\ i & 0 \end{pmatrix},\ \mathsf{u}_2 = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix},\ \mathsf{u}_3 = \begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix} \tag{40}$$

and for the Lie algebra $\mathfrak{so}(3)$ generators of the $SO(3)$ group

$$\mathsf{H}_1 = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{pmatrix},\quad \mathsf{H}_2 = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ -1 & 0 & 0 \end{pmatrix},$$
$$\mathsf{H}_3 = \begin{pmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}.$$

Other representations of $\mathfrak{su}(2)$ are possible and standard in Physics, with the algebra structure (39) changing slightly to $[\sigma_i, \sigma_j]_c = 2i\varepsilon_{ijk}\sigma_k$, which introduces the Pauli matrices:

$$\sigma_1 = -i\mathsf{u}_1\,,\quad \sigma_2 = i\mathsf{u}_2\,,\quad \sigma_3 = -i\mathsf{u}_3\,. \tag{41}$$

Regardless of the isomorphic $\mathfrak{su}(2)$ representations, the generated group $SU(2)$ is not isomorphic to $SO(3)$: $SU(2)$ is the simply connected double cover of $SO(3)$. To see this explicitly through the Rodrigues formulas, let

$$\mathsf{u} = \hat{\mathsf{u}} \cdot \mathbf{r} = r_i\,\mathsf{u}_i\,,\qquad \Theta = |\mathbf{\Omega}|\,,\qquad \mathbf{r} = \mathbf{\Omega}/\Theta\,. \tag{42}$$

Since $\mathsf{u}^2 = -\mathsf{u}_0$ for any unit vector $\mathbf{r}$, the matrix exponential in $SU(2)$ truncates to

$$U(\mathbf{\Omega}) = \exp\!\Big(\tfrac{1}{2}\,\hat{\mathsf{u}} \cdot \mathbf{\Omega}\Big) = \mathsf{u}_0 \cos\frac{\Theta}{2} + \mathsf{u}\,\sin\frac{\Theta}{2}\,. \tag{43}$$

The adjoint action of U on the pure-imaginary quaternion subspace induces a rotation $\mathsf{R}(\mathbf{\Omega}) \in SO(3)$:

$$U(\mathbf{\Omega})\ (\hat{\mathsf{u}} \cdot \mathbf{x})\ U(\mathbf{\Omega})^{-1} = \hat{\mathsf{u}} \cdot \big(\mathsf{R}(\mathbf{\Omega})\,\mathbf{x}\big)\,, \tag{44}$$

$$\mathsf{R}(\mathbf{\Omega})\,\mathbf{x} = \cos\Theta\ \mathbf{x} + (1-\cos\Theta)(\mathbf{r}\cdot\mathbf{x})\,\mathbf{r} + \sin\Theta\ \mathbf{r}\times\mathbf{x}\,. \tag{45}$$

In matrix form this is precisely the Rodrigues formula (36).

Equivalently, writing U in quaternion parametrisation,

$$U = q_0\,\mathsf{u}_0 + q_i\,\mathsf{u}_i\,,\qquad q_0^2 + \mathbf{q}^\top\mathbf{q} = 1\,,$$
$$q_0 = \cos\frac{\Theta}{2}\,,\qquad \mathbf{q} = \sin\frac{\Theta}{2}\,\mathbf{r}\,, \tag{46}$$

the induced rotation matrix becomes

$$\mathsf{R}(U) = (q_0^2 - \mathbf{q}^\top\mathbf{q})\,\mathsf{H}_0 + 2\,\mathbf{q}\,\mathbf{q}^\top + 2\,q_0\,[\mathbf{q}]_\times\,. \tag{47}$$

This exhibits $SU(2) \cong S^3$ directly.

a) The covering map.: The surjective homomorphism $\pi\colon SU(2) \to SO(3)$ is therefore given by

$$\pi(U) = \mathsf{R}(U)\,,\qquad \ker\pi = \{\pm\mathsf{u}_0\}\,. \tag{48}$$

Hence

$$SO(3) \cong SU(2)/\{\pm\mathsf{u}_0\} \cong S^3/\mathbb{Z}_2\,. \tag{49}$$

b) The two-to-one identification.: In the exponential parametrisation the double cover appears as

$$U(\mathbf{\Omega} + 2\pi\mathbf{r}) = -\,U(\mathbf{\Omega})\,,\qquad \pi(U) = \pi(-U)\,. \tag{50}$$

Antipodal points in $SU(2) \cong S^3$ therefore represent the same rotation in $SO(3)$. This is the precise topological origin of the phase-reflection phenomenon encountered in the previous subsection: the identification $(\mathbf{r}, \pi) \equiv (-\mathbf{r}, \pi)$ on the boundary of the Rodrigues parameter sphere is the quotient $S^3/\mathbb{Z}_2$ restricted to the equator. Equivalently, $\pi_1(SO(3)) = \mathbb{Z}_2$, whereas $SU(2) \cong S^3$ is simply connected.

c) Inverse map.: Given $\mathsf{R} \in SO(3)$, the rotation angle follows from $\operatorname{Tr}\mathsf{R} = 1 + 2\cos\Theta$ and the axis from the antisymmetric part, $n_k = -\varepsilon_{ijk}\,R_{ij}/(2\sin\Theta)$, after which (43) returns the $SU(2)$ element up to an overall sign. Alternatively, reading off from (47) directly: $q_0 = \pm\frac{1}{2}\sqrt{1+\operatorname{Tr}\mathsf{R}}$ and $q_k = -\varepsilon_{ijk}\,R_{ij}/(4q_0)$ for $q_0 \neq 0$. The sign ambiguity is precisely the $\mathbb{Z}_2$ kernel: no continuous global section $SO(3) \to SU(2)$ exists.

Frequently, we obtain a factorized rotation representation,

$$\exp(\Theta_c\mathsf{H}_c) = \exp(\Theta_a\mathsf{H}_a)\exp(\Theta_b\mathsf{H}_b), \tag{51}$$

presented with notation (37). An exponential series expansion of the right hand side of (51) is ordered with H_a to the left and H_b to the right. It is clear that a simple sum $\mathsf{H}_c \neq \mathsf{H}_a + \mathsf{H}_b$ cannot hold because its exponential series expansion is unordered. The Baker-Campbell-Hausdorff (BCH) formula[3] is useful to derive an additive relationship,

$$\begin{aligned}\Theta_c\mathsf{H}_c &= \Theta_b\mathsf{H}_b + \int_0^{\Theta_a} h\left(e^{s\,\mathrm{ad}(\mathsf{H}_a)}e^{\Theta_b\,\mathrm{ad}(\mathsf{H}_b)}\right)\mathsf{H}_a\,ds,\\ &= \Theta_a\mathsf{H}_a + \Theta_b\mathsf{H}_b + \frac{1}{2}\Theta_a\Theta_b[\mathsf{H}_a, \mathsf{H}_b] + \dots\,.\end{aligned} \tag{52}$$

However, (52) only hands an infinite series expansion. It is quite useful to have a closed expression for the Rodrigues vector $\mathbf{\Omega}_c$ of the resulting rotation $\mathsf{R}_c = \exp(\hat{\mathbf{H}} \cdot \mathbf{\Omega}_c)$. We could compactly denote this operation as $\mathbf{\Omega}_c = \ln \mathsf{R}_a\mathsf{R}_b\,.$

[3]The BCH formula can be derived using the derivative of exponentials, the resulting integral with function $h(x) = \ln(x)/(x-1)$ is understood as a power series expansion in x.

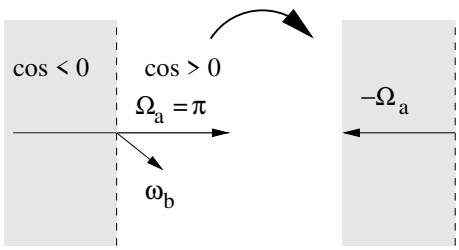

Fig. 7. Phase reflection occurs for $|\boldsymbol{\Omega}_a| = \pi$. The result of adding another infinitesimal vector $\boldsymbol{\omega}_b$ depends whether it has the same direction $\cos\vartheta > 0$, which will give a resulting vector in the direction $-\boldsymbol{\Omega}_a$. If $\cos\vartheta < 0$, the resulting vector will be unchanged.

Using the quaternion representation of $SU(2)$ developed in §B, one can show that the following Eqn. holds,

$$\boldsymbol{\Omega}_c = \frac{\Theta_c}{C}\left(a\,\mathbf{r}_a + b\,\mathbf{r}_b + ab\,\mathbf{r}_a \times \mathbf{r}_b\right), \qquad a = \tan\frac{\Theta_a}{2},\ b = \tan\frac{\Theta_b}{2}. \quad (53)$$

The normalizing factor, $C = \sqrt{A - \varepsilon^2}$, can be given as

$$A = 1 + a^2 + b^2 + a^2 b^2, \quad \varepsilon = (1 - ab\cos\vartheta), \quad \cos\vartheta = \mathbf{r}_a \cdot \mathbf{r}_b\ , \quad (54)$$

and the norm Θ_c

$$\tan\frac{\Theta_c}{2} = C/\,\varepsilon\ . \quad (55)$$

For the trivial case, $\cos\vartheta = \pm 1$ in (54), we simply obtain $\boldsymbol{\Omega}_c = \boldsymbol{\Omega}_a + \boldsymbol{\Omega}_b$. To evaluate the prefactor Θ_c/C in (53) or the primed versions below, we have to pay attention to the following two limit cases:

1) limit $\varepsilon \to 0$, (this implies $A \geq 1$)

$$\frac{\Theta_c}{C} = \frac{\pi}{\sqrt{A}}\left[1 - \frac{2}{\pi}\frac{\varepsilon}{\sqrt{A}} + \frac{1}{2}\left(\frac{\varepsilon}{\sqrt{A}}\right)^2 + \mathcal{O}((\varepsilon/\sqrt{A})^3)\right] \quad (56)$$

2) limit $C \to 0$, (this implies $\varepsilon^2 \geq 1$)

$$\frac{\Theta_c}{C} = \frac{2}{\varepsilon}\left[1 - \frac{1}{3}\left(\frac{C}{\varepsilon}\right)^2 + \mathcal{O}((C/\varepsilon)^4)\right]. \quad (57)$$

C. Phase reflection at $\Theta = \pi$

Since the logarithmic map loses its uniqueness at $\Theta = \pi$ (cf. the double-cover discussion in §B), we have to be careful to define the vector addition (53). Firstly, we consider $\Theta_a \to \pi-$ which implies $a \to \infty$ and b finite. Secondly a similar statement accounts for the limit $a \to \infty$ and $b \to \infty$. We notice that those two limits, (i) and (ii), can be treated consistently. In both cases a cutline at $\cos\vartheta = 0$ has to be made to account for the phase reflection boundary. The limit process $\Theta_a \to \pi-$ can be depicted graphically, see Fig. 7. Hence, all quantities are properly defined in the limits. Therefore, Eqn. (53) can be used to add finite Rodrigues vectors without employing intermediate rotation matrices or quaternions.

D. Adding two infinitesimal rotations: kinematics of rigid bodies

The rate of change, angular velocity, of $\boldsymbol{\Omega}$ for an infinitesimal small quantity $dt \to 0$ is $d\boldsymbol{\Omega} = dt\,\boldsymbol{\omega}$. We can apply the finite addition formula (53) or directly apply the BCH expansion (52) to obtain,

$$d\boldsymbol{\Omega}_c = dt\,\boldsymbol{\omega}_c = dt\,\boldsymbol{\omega}_a + dt\,\boldsymbol{\omega}_b + \frac{1}{2}dt^2\,\boldsymbol{\omega}_a \times \boldsymbol{\omega}_b\ . \quad (58)$$

As such we recover in (58) the well known fact that angular velocities $\boldsymbol{\omega}$ constitute an algebraic vector space with the normal commutative addition. More exactly, the vector space is known as $T_1SO(3)$.

E. Right Rodrigues Jacobian

We now derive the right Jacobian J_{ω_r} that relates infinitesimal perturbations in Lie-algebra coordinates to the corresponding group-level variations. We present two routes: first via the Wei-Norman formula using the derivative of exponentials, then a direct computation from the finite addition formula (53).

We consider the rotation product expression in (51). We describe the infinitesimal environment at $\boldsymbol{\Omega}_c = \boldsymbol{\Omega}_a + \Delta\boldsymbol{\omega}$ for a finite left rotation $\boldsymbol{\Omega}_a$ in dependence of an infinitesimal right rotation $\boldsymbol{\Omega}_b \to \mathbf{0}$. Then we have to analyze the following directional derivative

$$\frac{d}{ds}\ \exp(\hat{\mathbf{H}}\cdot\boldsymbol{\Omega}_a + s\,\hat{\mathbf{H}}\cdot\Delta\boldsymbol{\omega})\ =\ \exp(\hat{\mathbf{H}}\cdot\boldsymbol{\Omega}_a)\hat{\mathbf{H}}\cdot\mathsf{J}_{\omega_r}^{-1}(\boldsymbol{\Omega}_a)\Delta\boldsymbol{\omega}, \quad (59)$$

where the right composition Jacobian $\mathsf{J}_{\omega_r}(\boldsymbol{\Omega}_a) \in \mathbb{R}^{3\times 3}$: $T_{\mathsf{R}(\boldsymbol{\Omega}_a)}SO(3) \to T_1SO(3)$ describes the mapping from the tangent space at $\mathsf{R}(\boldsymbol{\Omega}_a)$ to the one at identity. In order to evaluate the left hand side of (59), it is advantageous to consider a "matrix" equation for $g \in SE(3)$ and general $\mathsf{A}, \mathsf{B} \in \mathfrak{se}(3)$

$$\frac{\partial}{\partial t}g(t,s) = (\mathsf{A} + s\mathsf{B})g(s,t)\ \to\ g(s,t) = e^{t(\mathsf{A}+s\mathsf{B})}. \quad (60)$$

With the help of the Dyson ordering operator $\mathcal{T}$ and the adjoint notation (32), it can be shown that the following holds

$$g(s,t) = e^{t\mathsf{A}}\,\mathcal{T}\exp\left(s\int_0^t e^{-u\,\mathrm{ad}(\mathsf{A})}\,du\ \mathsf{B}\right)\ . \quad (61)$$

Differentiating (61) with respect to s, setting $t = 1$, then applying it to (59) yields

$$u(\Theta_a\,\mathrm{ad}(\mathsf{H}_a))\hat{\mathbf{H}}\cdot\Delta\boldsymbol{\omega} = \hat{\mathbf{H}}\cdot\mathsf{J}_{\omega_r}^{-1}(\boldsymbol{\Omega}_a)\Delta\boldsymbol{\omega}, \quad u(x) = \int_0^1 e^{-sx}\,ds \quad (62)$$

a relationship to obtain $\mathsf{J}_{\omega_r}(\boldsymbol{\Omega}_a)$. In fact (62) constitutes a Wei-Norman formula [12], which relates the differential changes of a product of exponentials (51) to the changes of one resulting exponential. However, it is not easy to obtain a closed form expression from (62).

a) Direct route via finite addition.: It shall be noticed that with Eqn. (53) we directly obtain the right Rodrigues Jacobian $\mathsf{J}_{\omega_r}(\boldsymbol{\Omega}_a)$ by,

$$\mathsf{J}_{\omega_r}(\boldsymbol{\Omega}_a) = \left.\frac{\partial \boldsymbol{\Omega}_c}{\partial \boldsymbol{\Omega}_b}\right|_{\Theta_b=0} = \left.\frac{\partial \boldsymbol{\Omega}_c}{\partial \Theta_b}\right|_{\Theta_b=0} \mathbf{r}_b + \left.\frac{\partial \boldsymbol{\Omega}_c}{\partial \mathbf{r}_b}\frac{\partial \mathbf{r}_b}{\partial \boldsymbol{\Omega}_b}\right|_{\Theta_b=0}. \tag{63}$$

Further, Eqn. (63) can be evaluated as,

$$\mathsf{J}_{\omega_r}(\boldsymbol{\Omega}_a) = (1-x)\,\mathsf{I} + x\,\mathbf{r}_a\mathbf{r}_a^T + \frac{1}{2}[\boldsymbol{\Omega}_a]_\times,\ 1-x = \frac{\Theta_a}{2}\tan^{-1}\frac{\Theta_a}{2}. \tag{64}$$

Similarly, we could leave $\boldsymbol{\Omega}_b$ in the product expansion (51) at a finite value and consider infinitesimal left variations of $\boldsymbol{\Omega}_a \to \mathbf{0}$. This naturally leads to the complementary left Rodrigues Jacobian

$$\mathsf{J}_{\omega_l}(\boldsymbol{\Omega}_b) = (1-y)\,\mathsf{I} + y\,\mathbf{r}_b\mathbf{r}_b^T - \frac{1}{2}[\boldsymbol{\Omega}_b]_\times,\ 1-y = \frac{\Theta_b}{2}\tan^{-1}\frac{\Theta_b}{2}. \tag{65}$$

It shall be emphasised that the Jacobians above don't have singularities and are invertible, since all quantities are well defined for $\Theta \leq \pi$ and $\det(\mathsf{J}_{\omega_r}) = (1-x)^2 + \Theta_a^2/4$.

F. $SE(3)$ Composition Jacobians

In close analogy to the previous sections, a composition in $SE(3)$ can be analyzed in the additive framework,

$$(\mathsf{R}(\boldsymbol{\Omega}_c), \mathbf{T}_c) = (\mathsf{R}_a, \mathbf{T}_a)\cdot(\mathsf{R}_b, \mathbf{T}_b)\,. \tag{66}$$

1) Composing finite motions: With the group composition (2), we can formally write

$$\ln(\mathsf{R}(\boldsymbol{\Omega}_c), \mathbf{T}_c) = [\ln \mathsf{R}_a\mathsf{R}_b,\ \mathsf{S}^{-1}(\boldsymbol{\Omega}_c)(\mathsf{R}_a\mathbf{T}_b + \mathbf{T}_a)\,]. \tag{67}$$

2) Composing a finite with an infinitesimal motion: The addition formula (67) can be analyzed in the limit $t \to 0$ where the right composition is $\ln(\mathsf{R}_b, \mathbf{T}_b) = [t\boldsymbol{\omega}_b, t\mathbf{v}_b]$. It is easy to show that the following must hold to order $\mathcal{O}(t)$,

$$\begin{aligned}[\boldsymbol{\Omega}_c, \mathbf{t}_c] &= [\boldsymbol{\Omega}_a, \mathbf{t}_a] + \\ &[\mathsf{J}_{\omega_r}(\boldsymbol{\Omega}_a)\boldsymbol{\omega}_b,\ \mathsf{S}^{-1}(\boldsymbol{\Omega}_a)\mathsf{R}(\boldsymbol{\Omega}_a)\mathbf{v}_b + \mathsf{J}_{t_r}(\boldsymbol{\Omega}_a, \mathbf{t}_a)\boldsymbol{\omega}_b].\end{aligned} \tag{68}$$

3) The S^{-1} identity and coupling Jacobian: In (36) we only gave the expression for S. We can use the properties of the finite H algebra with its characteristic polynomial $\mathsf{H}(\mathsf{H}^2 + \mathsf{I}) = 0$ to obtain an expression for $\mathsf{S}^{-1}(\boldsymbol{\Omega}_a)$. As an interesting fact, we meet the Rodrigues Jacobians again

$$\begin{aligned}\mathsf{S}^{-1}(\boldsymbol{\Omega}_a) &= (1-x)\mathsf{I} + x\,\mathbf{r}_a\mathbf{r}_a^T - \frac{1}{2}[\boldsymbol{\Omega}_a]_\times \\ &= \mathsf{J}_{\omega_r}(-\boldsymbol{\Omega}_a) = \mathsf{J}_{\omega_l}(\boldsymbol{\Omega}_a)\,.\end{aligned} \tag{69}$$

Further, one can show that

$$\mathsf{S}^{-1}(\boldsymbol{\Omega}_a)\mathsf{R}(\boldsymbol{\Omega}_a) = \mathsf{R}(\boldsymbol{\Omega}_a)\mathsf{S}^{-1}(\boldsymbol{\Omega}_a) = \mathsf{J}_{\omega_r}(\boldsymbol{\Omega}_a), \tag{70}$$

A more detailed calculation for the coupling Jacobian J_{t_r} yields

$$\mathsf{J}_t(\boldsymbol{\Omega}_a, \mathbf{t}_a) = \frac{\partial[\mathsf{S}^{-1}(\boldsymbol{\Omega}_a)\mathbf{T}_a]}{\partial \boldsymbol{\Omega}_a}\,\mathsf{J}_{\omega_r}(\boldsymbol{\Omega}_a), \quad \mathbf{T}_a = \mathsf{S}(\boldsymbol{\Omega}_a)\,\mathbf{t}_a. \tag{71}$$

We can show that

$$\begin{aligned}\frac{\partial[\mathsf{S}^{-1}(\boldsymbol{\Omega}_a)\mathbf{T}_a]}{\partial \boldsymbol{\Omega}_a} &= \frac{1}{2}[\mathbf{t}_a]_\times + \frac{x}{\Theta_a}\left(\mathbf{r}_a\mathbf{t}_a^T - t_a\,\mathsf{H}_a^2 + \mathbf{t}_a\mathbf{r}_a^T\right) + \\ &\quad \frac{t_a}{\Theta_a}(\mathsf{S}(\boldsymbol{\Omega}_a) - \mathsf{S}^{-1}(-\boldsymbol{\Omega}_a)),\end{aligned} \tag{72}$$

with $t_a = \mathbf{r}_a\cdot\mathbf{t}_a$, which finishes the discourse on the $SE(3)$ composition Jacobians.

G. Additive versus compositive derivative

The Rodrigues formula (36) allows differentiation with respect to $\boldsymbol{\Omega}$ directly,

$$\mathsf{R}(\boldsymbol{\Omega}_0 + \Delta\boldsymbol{\omega}) \approx \mathsf{R}(\boldsymbol{\Omega}_0) + \left.\frac{\partial \mathsf{R}}{\partial \Omega_\mu}\right|_{\boldsymbol{\Omega}_0}\Delta\omega_\mu = \mathsf{R}(\boldsymbol{\Omega}_0)(\mathsf{I} + [\delta\boldsymbol{\Omega}]_\times). \tag{73}$$

We can call $\partial\mathsf{R}/\partial\Omega_\mu$ the additive derivative, whereas $[\delta\boldsymbol{\Omega}]_\times = \left.\frac{d}{dt}\right|_{t=0}\exp(t[\delta\boldsymbol{\Omega}]_\times)$ can be termed the compositive (incremental) derivative.

In accordance with the vector addition over $SO(3)$ (53), we can show that

$$\delta\boldsymbol{\Omega} \;=\; \mathsf{J}_{\omega_r}^{-1}(\boldsymbol{\Omega}_0)\,\Delta\boldsymbol{\omega} \tag{74}$$

or explicitly

$$\delta\boldsymbol{\Omega} = \frac{\sin\Theta_0}{\Theta_0}\Delta\boldsymbol{\omega} + \frac{1-\cos\Theta_0}{\Theta_0}\,\Delta\boldsymbol{\omega}\times\mathbf{r}_0 + k\,(1 - \frac{\sin\Theta_0}{\Theta_0})\,\mathbf{r}_0$$

with $k = \mathbf{r}_0\cdot\Delta\boldsymbol{\omega}$ and the usual definition $\Theta_0 = |\boldsymbol{\Omega}_0|$, $\mathbf{r}_0 = \boldsymbol{\Omega}_0/\Theta_0$.

We derive the second-order corrections to mean and covariance for the composition $h = g_0\cdot\Delta g\cdot f_0\cdot\Delta f$, where $\Delta g = \exp([\boldsymbol{\omega}_g, \mathbf{v}_g])$ and $\Delta f = \exp([\boldsymbol{\omega}_f, \mathbf{v}_f])$ are zero-mean perturbations with covariances Σ_{gg} and Σ_{ff} respectively.

H. Mean correction from BCH curvature

Using the finite composition formula (53) rather than the truncated BCH series, the composed rotation in exponential coordinates can be expanded to second order as

$$\boldsymbol{\Omega}_c = \boldsymbol{\Omega}_a + \boldsymbol{\Omega}_b + \frac{1}{2}\boldsymbol{\Omega}_a\times\boldsymbol{\Omega}_b + \frac{1}{12}\left(\boldsymbol{\Omega}_a\times(\boldsymbol{\Omega}_a\times\boldsymbol{\Omega}_b) + \boldsymbol{\Omega}_b\times(\boldsymbol{\Omega}_b\times\boldsymbol{\Omega}_a)\right) + \cdot \tag{75}$$

Taking the expectation with $\langle\boldsymbol{\Omega}_a\rangle = 0$, $\langle\boldsymbol{\Omega}_b\rangle = 0$:

$$\langle\boldsymbol{\Omega}_c\rangle = \frac{1}{2}\langle\boldsymbol{\Omega}_a\times\boldsymbol{\Omega}_b\rangle = \frac{1}{2}\sum_{\alpha\beta}\varepsilon_{\cdot\alpha\beta}\,\langle\omega_{a,\alpha}\,\omega_{b,\beta}\rangle. \tag{76}$$

For the translational component in $SE(3)$, the coupling through S^{-1} introduces additional terms. Using (68) expanded to second order, the full $SE(3)$ mean correction is

$$\delta\mu^{(2)} = \frac{1}{2}\begin{pmatrix}\varepsilon_{i\alpha\beta}\,[\Sigma_{gf}]^{\omega\omega}_{\alpha\beta} \\ \varepsilon_{i\alpha\beta}\,[\Sigma_{gf}]^{\omega v}_{\alpha\beta} + [\mathsf{Q}(\boldsymbol{\Omega}_0)]_{ij}\,[\Sigma_{ff}]^{\omega\omega}_{j}\end{pmatrix}, \tag{77}$$

where $\mathsf{Q}(\boldsymbol{\Omega}_0)$ encodes the Hessian of $\mathsf{S}^{-1}(\boldsymbol{\Omega})\mathsf{R}(\boldsymbol{\Omega})$ evaluated at the base point, and the superscripts denote the rotational (ω) and translational (v) blocks of the cross-covariance.

I. Covariance correction

The second-order covariance correction arises from the quadratic term $Q_i = \frac{1}{2}\varepsilon_{i\alpha\beta}\,\omega_{a,\alpha}\,\omega_{b,\beta}$ in the BCH expansion. Writing $\boldsymbol{\Omega}_c = L + Q + \cdots$ where $L_i = \omega_{a,i} + \omega_{b,i}$ is the linear part, the covariance picks up a correction from $\langle Q_i Q_j\rangle - \langle Q_i\rangle\langle Q_j\rangle$.

1) Rotational block (3×3): Since Q_i mixes components from two distinct random vectors $\boldsymbol{\omega}_a$ (from Δg) and $\boldsymbol{\omega}_b$ (from Δf), the indices must be tracked separately:

$$\Delta\Sigma_{ij}^{(2),\omega\omega} = \frac{1}{4}\sum_{\alpha\beta\gamma\delta}\varepsilon_{i\alpha\beta}\,\varepsilon_{j\gamma\delta}\,\mathcal{C}_{\alpha\beta\gamma\delta}, \tag{78}$$

where the fourth-moment tensor is

$$\mathcal{C}_{\alpha\beta\gamma\delta} = \langle\omega_{a,\alpha}\,\omega_{b,\beta}\,\omega_{a,\gamma}\,\omega_{b,\delta}\rangle - \langle\omega_{a,\alpha}\,\omega_{b,\beta}\rangle\langle\omega_{a,\gamma}\,\omega_{b,\delta}\rangle. \tag{79}$$

For Gaussian perturbations, the fourth-order cumulant vanishes identically. The correction comes entirely from the factorization of the fourth moment via Isserlis' theorem:

$$\begin{aligned}\langle\omega_{a,\alpha}\,\omega_{b,\beta}\,\omega_{a,\gamma}\,\omega_{b,\delta}\rangle &= [\Sigma_{aa}]_{\alpha\gamma}\,[\Sigma_{bb}]_{\beta\delta}\\ &\quad + [\Sigma_{ab}]_{\alpha\delta}\,[\Sigma_{ba}]_{\beta\gamma} + [\Sigma_{ab}]_{\alpha\beta}\,[\Sigma_{ab}]_{\gamma\delta}.\end{aligned} \tag{80}$$

The last term cancels with the subtracted product, giving the closed-form correction:

$$\boxed{\Delta\Sigma_{ij}^{(2),\omega\omega} = \frac{1}{4}\sum_{\alpha\beta\gamma\delta}\varepsilon_{i\alpha\beta}\,\varepsilon_{j\gamma\delta}\Big([\Sigma_{aa}]_{\alpha\gamma}\,[\Sigma_{bb}]_{\beta\delta} + [\Sigma_{ab}]_{\alpha\delta}\,[\Sigma_{ba}]_{\beta\gamma}\Big)} \tag{81}$$

When the perturbations are independent ($\Sigma_{ab} = 0$), this simplifies to

$$\Delta\Sigma_{ij}^{(2),\omega\omega} = \frac{1}{4}\varepsilon_{i\alpha\beta}\,\varepsilon_{j\gamma\delta}\,[\Sigma_{aa}]_{\alpha\gamma}\,[\Sigma_{bb}]_{\beta\delta}, \tag{82}$$

which can be evaluated using the identity $\varepsilon_{i\alpha\beta}\varepsilon_{j\gamma\delta}A_{\alpha\gamma}B_{\beta\delta} = (\operatorname{tr}\mathsf{A})(\operatorname{tr}\mathsf{B})\delta_{ij} - (\operatorname{tr}\mathsf{A})B_{ij} - A_{ij}(\operatorname{tr}\mathsf{B}) + A_{ik}B_{kj} + A_{jk}B_{ki}$.

2) Rotation-translation cross block (3×3): The translational component $t_{c,i}$ receives a second-order contribution through the coupling $\mathsf{S}^{-1}(\boldsymbol{\Omega})\mathsf{R}(\boldsymbol{\Omega})$, expanded around the base point. Using the SE(3) vector addition (68), the cross-covariance correction is

$$\begin{aligned}\Delta\Sigma_{ij}^{(2),\omega t} = \frac{1}{2}\varepsilon_{i\alpha\beta}\Big([\Sigma_{aa}]_{\alpha\gamma}\,[\mathsf{J}_{\omega_r}]_{\gamma k}\,[\Sigma_{bb}^{vv}]_{\beta j}\\ + [\Sigma_{ab}^{\omega v}]_{\alpha j}\,[\Sigma_{ba}^{\omega\omega}]_{\beta\gamma}\,[\mathsf{J}_t]_{\gamma k}\Big) + \mathcal{O}(\sigma^5),\end{aligned} \tag{83}$$

where $[\Sigma^{vv}]$, $[\Sigma^{\omega v}]$ denote the translational and cross blocks of the 6×6 covariance, and J_t is the coupling Jacobian from (71).

3) Translational block (3×3): The pure translational correction receives contributions from: (i) the J_t coupling acting on the rotational perturbation, and (ii) the Hessian of $\mathsf{S}^{-1}\mathsf{R}$ at the base point. To leading order in σ^4:

$$\begin{aligned}\Delta\Sigma_{ij}^{(2),tt} &= [\mathsf{Q}]_{ik}\,[\Sigma_{aa}^{\omega\omega}]_{kl}\,[\mathsf{Q}]_{lj}^T\\ &\quad + \frac{1}{4}\varepsilon_{i'\alpha\beta}\,\varepsilon_{j'\gamma\delta}\,[\mathsf{J}_t]_{ii'}\,[\mathsf{J}_t]_{jj'}\,\mathcal{C}_{\alpha\beta\gamma\delta} + \mathcal{O}(\sigma^5),\end{aligned} \tag{84}$$

where $\mathsf{Q}(\boldsymbol{\Omega}_0)$ is the Hessian matrix from the mean correction (77) and $\mathcal{C}_{\alpha\beta\gamma\delta}$ is the same Isserlis-reduced tensor as in (81).

Remark 3. The key insight is that for Gaussian perturbations, no higher-order moments are needed—the entire correction is determined by the second-order covariance blocks Σ_{aa}, Σ_{bb}, Σ_{ab}, combined with the structure constants ε_{ijk} and the coupling Jacobian J_t. The correction vanishes on commutative groups (where $\varepsilon_{ijk} = 0$) and on $SO(3)$ only the rotational block (81) survives.

J. Moment generating function for projective observations

For a single landmark $\mathbf{x}$ observed from camera f with projective model $\pi(f\star\mathbf{x}) = [u, v, 1]$ where $u = x_1'/x_3'$, $v = x_2'/x_3'$, the log-likelihood as a function of $\mathbf{x}$ (with f fixed) is

$$\ell(\mathbf{x}) = -\frac{1}{2}(\mathbf{z} - \pi(f\star\mathbf{x}))^T\Sigma_{zz}^{-1}(\mathbf{z} - \pi(f\star\mathbf{x})). \tag{85}$$

The moment generating function of the integrand in the marginalization (22) is

$$K(\mathbf{s}) = \ell(\mathbf{x}_{\text{opt}}) + \frac{1}{2}\mathbf{s}^T\mathsf{H}^{-1}\mathbf{s} + \frac{1}{6}\sum_{abc}\kappa_{abc}\,s_a s_b s_c + \cdots \tag{86}$$

where $\mathsf{H} = -\nabla^2\ell|_{\mathbf{x}_{\text{opt}}}$ is the Hessian (positive definite at the optimum) and κ_{abc} are the third cumulants.

K. Third cumulants of the projective model

The third derivatives of π at $\mathbf{x}' = f\star\mathbf{x}$ are:

$$\frac{\partial^3 u}{\partial x_a'\partial x_b'\partial x_c'} = \frac{\partial^3}{\partial x_a'\partial x_b'\partial x_c'}\left(\frac{x_1'}{x_3'}\right), \tag{87}$$

which are rational functions with poles at $x_3' = 0$ (behind the camera). The non-vanishing components are:

$$\begin{aligned}\frac{\partial^3 u}{\partial x_1'\partial x_3'^2} &= \frac{2}{x_3'^3}, \quad \frac{\partial^3 u}{\partial x_3'^3} = \frac{6\,u}{x_3'^3},\\ \frac{\partial^3 v}{\partial x_2'\partial x_3'^2} &= \frac{2}{x_3'^3}, \quad \frac{\partial^3 v}{\partial x_3'^3} = \frac{6\,v}{x_3'^3}.\end{aligned} \tag{88}$$

The third cumulants in the original landmark coordinates $\mathbf{x}$ are obtained by chain rule through R_f:

$$\kappa_{abc} = \sum_{a'b'c'}\frac{\partial^3\ell}{\partial x_{a'}'\partial x_{b'}'\partial x_{c'}'}\,R_{a'a}\,R_{b'b}\,R_{c'c}. \tag{89}$$

L. Saddlepoint correction to Laplace

The saddlepoint approximation to the marginal contribution of landmark j is obtained by solving the saddlepoint equation $K'(\mathbf{s}_0) = 0$ and evaluating:

$$\int e^{\ell(\mathbf{x})}d\mathbf{x} \approx e^{\ell(\mathbf{x}_{\text{opt}})}\,(2\pi)^{3/2}|\mathsf{H}|^{-1/2}\left(1 + \frac{1}{8}\sum_{abcd}\kappa_{abcd}\,[\mathsf{H}^{-1}]_{ab}[\mathsf{H}^{-1}]_{cd} + \right. \tag{90}$$

where κ_{abcd} are the fourth cumulants and $[\mathsf{H}^{-3}]_{abc} = \sum_{def}[\mathsf{H}^{-1}]_{ad}[\mathsf{H}^{-1}]_{be}[\mathsf{H}^{-1}]_{cf}$.

The Laplace approximation retains only the leading term (the first factor in (90)). The correction in parentheses is $\mathcal{O}(1/n)$ for n observations and becomes significant when: (a) the depth x_3' is comparable to the baseline, (b) few cameras observe the landmark, or (c) the landmark is near the image boundary where u or v are large.

The corrected marginal log-posterior over camera poses is then

$$\ln p(f|\mathbf{z}) = \sum_j \left[\ell_j(\mathbf{x}^j_{\text{opt}}(f)) - \frac{1}{2} \ln |\mathsf{H}_j(f)| + \ln(1 + \delta^{\text{SP}}_j(f)) \right] + \ln p(f), \tag{91}$$

where δ^{SP}_j is the saddlepoint correction term from (90). This expression is optimized over $f \in SE(3)$ using the Gauss-Newton iteration with compositive updates $f_{t+1} = f_t \cdot \exp(\Delta \mathbf{f}_t)$ as described in Section IV.

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